MATH 127 Calculus for the Sciences

Lecture 25

Today's lecture

Last time
Asymptotes
Intervals of Increase/Decrease

This time
Course note coverage Section 3.3.2 and 3.3.3
Critical points
Local extrema

Recap

We have a function $f(x)$, then we take the derivative and get

$$f'(x)$$

Now at a input $x = a$, we have

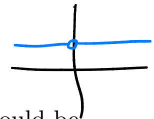
1. if $f'(a) > 0$, then $f(x)$ is $\boxed{\text{increasing}}$ at $x = a$;
2. if $f'(a) < 0$, then $f(x)$ is $\boxed{\text{decreasing}}$ at $x = a$;
3. if $f'(a) = 0$, then $f(x)$ is $\boxed{\text{a critical point}}$ at $x = a$;
4. if $f'(a)$ does not exist, then $f(x)$ is $\boxed{\text{not differentiable}}$ and $\boxed{\text{critical point.}}$ at $x = a$.

Recap

For horizontal asymptotes

1. Draw graph
2. If graph is too hard, take limit $x \to \infty$ and $x \to -\infty$.

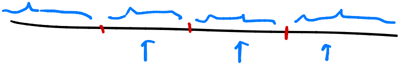
$f(x) = \frac{x}{x}$



For vertical asymptotes

1. Draw graph
2. If graph is too hard, list out where potential asymptotes could be
3. then take limits to check if they are indeed asymptotes

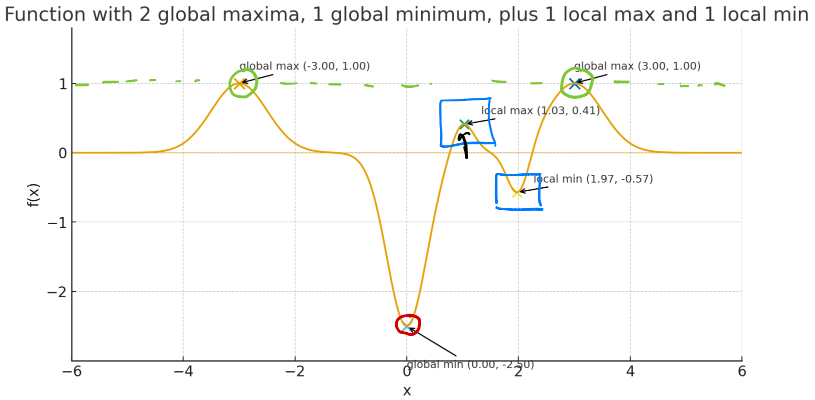
For intervals of increase/decrease



1. Draw graph
2. If graph is too hard, calculate derivative
3. Find where the critical points are, then split x-axis using those points
4. on each piece of the splitting, check the sign of derivative and apply derivative test.

Max and min

You have hear of max and min. Those are usually refered to as global max and global min. There is also a notion of local max and local min.



But how should we defined the notion of local max and local min in mathematics?

Max and min

Definition

- An input $x = a$ is a **global maximum** of f if

$$f(a) \geq \boxed{f(x)} \text{ for all } x \text{ in the domain of } f(x).$$

- An input $x = a$ is a **global minimum** of f if

$$f(a) \leq \boxed{f(x)} \text{ for all } x \text{ in the domain of } f(x).$$

Definition

- An input $x = a$ is a **local maximum** of f if

$f(a) \geq f(x)$ for all x in a neighborhood around a.

- An input $x = a$ is a **local minimum** of f if

$f(a) \leq f(x)$ for all x in a neighborhood around a.

Potential positions of local max and min

Question How do we check where a function has a local max or min? I don't want to manually check for $x = a$ for every single number a. There's infinitely many of them!

Theorem (Fermat's Theorem)
If $f(x)$ has a local maximum or local minimum at $x = a$, then

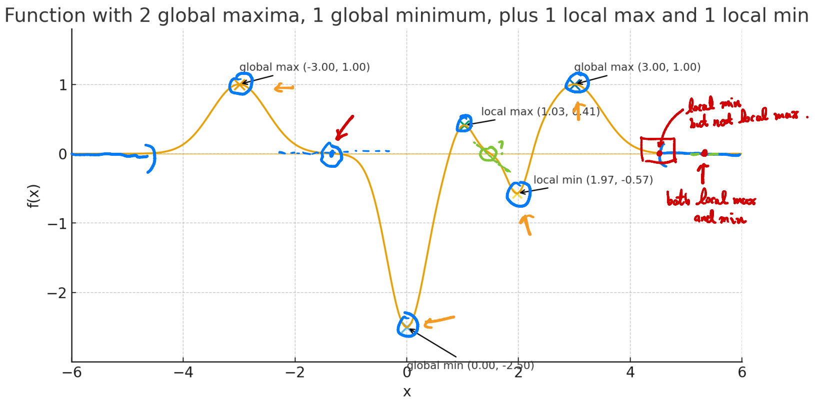
$$f(a) \text{ is a critical point, i.e}$$

$$f'(a) = \boxed{0} \quad \text{or} \quad f'(a) \ \boxed{\text{does not exist.}}$$

Conclusion To solve for the set of local max/min, you can first find all the critical points (which are potentially local max/min), and then you only need to check whether those points are local max/min.

Potential positions of local max and min

So we know all local max/min are critical. Which of the following points are critical but not local max/min?



Local max and min

Incorrect definition

- An input $x = a$ is a **local maximum** of f if

$$f(x) \text{ is increasing on the left of } a$$

$$f(x) \text{ is decreasing on the right of } a$$

- An input $x = a$ is a **local minimum** of f if

$$f(x) \text{ is decreasing on the left of } a$$

$$f(x) \text{ is increasing on the right of } a$$

Why is this definition incorrect, what situation does it miss?

Local max and min

Derivative test $\boxed{\text{Suppose } f(x) \text{ is continuous, then}}$

- An input $x = a$ is a **local maximum** of f if

$$f(x) \text{ is increasing on the left of } a \xleftrightarrow{\text{same as}} f'(x) > 0$$

$$f(x) \text{ is decreasing on the right of } a \xleftrightarrow{\text{same as}} f'(x) < 0$$

- An input $x = a$ is a **local minimum** of f if

$$f(x) \text{ is decreasing on the left of } a \longleftrightarrow \text{----.}$$

$$f(x) \text{ is increasing on the right of } a \longleftrightarrow \text{----.}$$

Question The statement says nothing about derivatives. Why is this called the derivative test?

Finding local max/min

Here are the steps you should use to find local max/min.

1. Compute critical points
2. Fermat's theorem says that local min/max can only happen at critical points.
3. If the function is continuous, then check the following at the critical points:
   3.1 If the derivative switches from negative to positive, that's a local minimum!
   3.2 If the derivative switches from positive to negative, that's a local maximum!
4. Otherwise, use definition
   4.1 Check whether the output of the function is higher near the critical points, if so, that's a local minimum!
   4.2 Check whether the output of the function is lower near the critical points, if so, that's a local maximum!

Finding local max/min

Example Find the local maxima and minima of the function

$$f(x) = x^2 \ln |x|$$

First look for critical points.

$$f'(x) = 2x\ln|x| + x$$
$$= x(2\ln|x| + 1)$$

$f'(x)$ is undefined at 0.
but 0 is not in the domain

$f'(x) = 0$ if

$$2\ln|x| + 1 = 0$$
$$\ln|x| = -\tfrac{1}{2}$$
$$|x| = e^{-\frac{1}{2}}$$
$$x = \pm e^{-\frac{1}{2}}$$

$-e^{-\frac{1}{2}}$ $\quad 0 \quad$ $e^{\frac{1}{2}}$

	$x < -e^{-\frac{1}{2}}$	$-e^{-\frac{1}{2}} < x < 0$	$0 < x < e^{\frac{1}{2}}$	$e^{\frac{1}{2}} < x$
x	$-$	$-$	$+$	$+$
$2\ln\lvert x\rvert+1$	$+$	$-$	$-$	$+$
$f'(x)$	$-$	$+$	$-$	$+$

$x = -e^{-\frac{1}{2}}$ is a local minimum

$x = +e^{-\frac{1}{2}}$ is a local minimum

Finding local max/min

Example Assume $f(x) = \int_0^x \frac{\sin(t)}{t}\,dt$ is continuous on $[0, 2\pi]$. Find the local min/max of $f(x)$ within the interval $[0, 2\pi]$.

Find critical points.

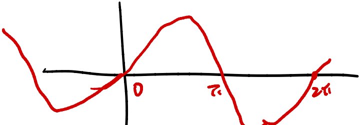
$f'(x) = \dfrac{\sin x}{x}$

$f'(x)$ is undefined at $x=0$

$f'(x) = 0$ when

$\sin x = 0$

$x = \pi$ or 2π.



	$x<0$	$0<x<\pi$	$\pi<x<2\pi$	$2\pi<x$
$\sin(x)$	−	+	−	+
x	−	+	+	+
$f'(x)$	+	+	−	+

$x=0$ is neither

$x=\pi$ is a local max

$x=2\pi$ is a local min

Finding local max/min

Example Make up a function with a jump discontinuity and has a local maximum at the discontinuity.
What about a function with a jump discontinuity but the discontinuity point is neither a local max or min?